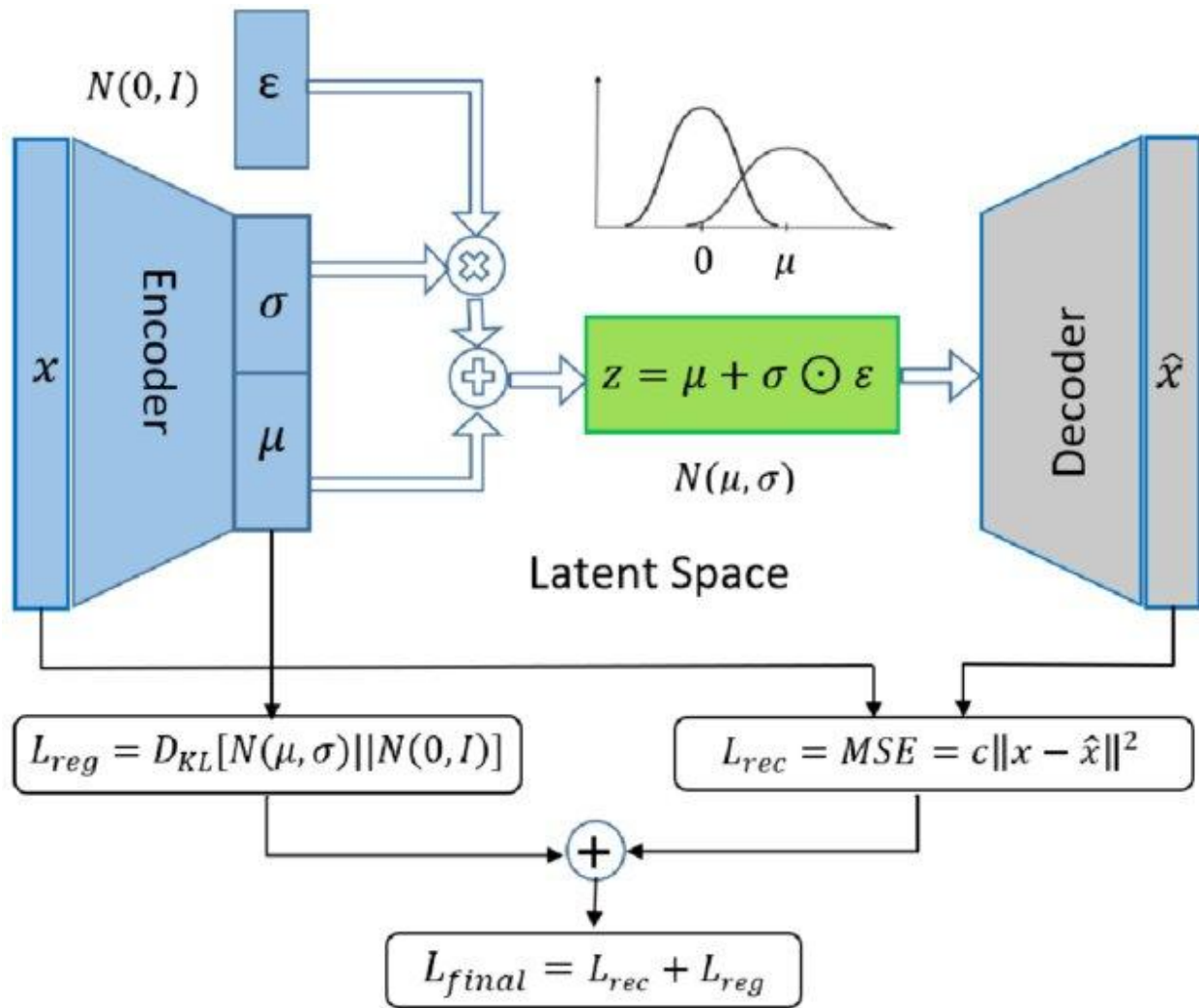
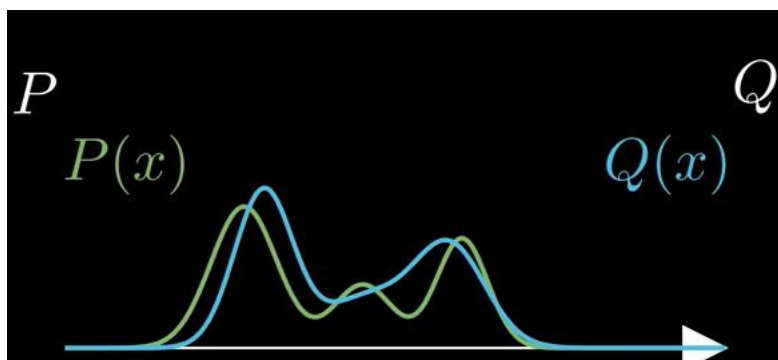


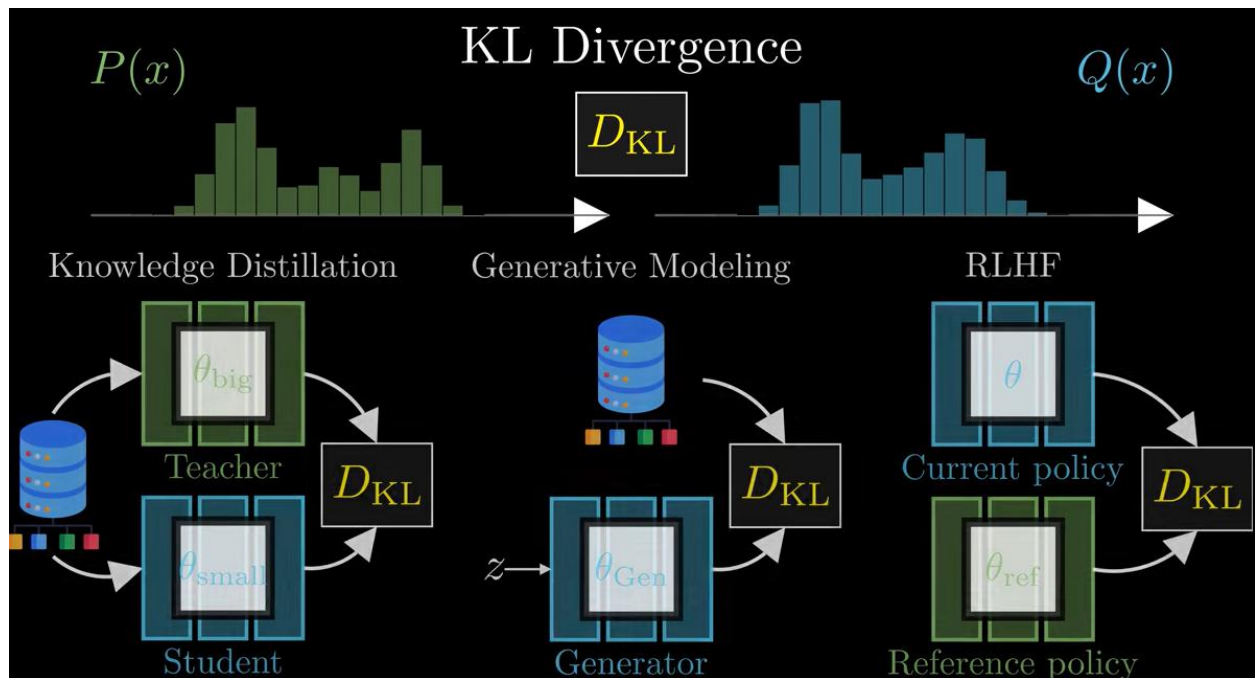
Feature	Standard Autoencoder (AE)	Variational Autoencoder (VAE)
Primary Purpose	Dimensionality Reduction (finding an efficient, compressed representation).	Generative Modeling (learning the probability distribution of data).
Latent Space Output	The Encoder outputs a single, deterministic vector for each input.	The Encoder outputs two vectors: a mean vector μ and a standard deviation vector σ defining a distribution.
Latent Space Structure	Discontinuous and Unstructured. Gaps exist between clusters that decode to noise/artifacts.	Continuous and Structured. The entire space is forced to follow a known distribution (typically Gaussian $N(0, 1)$).
Loss Function	Only Reconstruction Loss (e.g., Mean Squared Error)	Reconstruction Loss + Regularization Loss (KL Divergence).
KL Divergence	Not used.	Used to force the encoder's output distributions to align with a simple prior distribution.
Data Generation	Poor/Difficult. Randomly sampling from the latent space is likely to produce nonsense.	Excellent/Easy. Simply sample a point h from the standard Gaussian $N(0, 1)$ and feed it to the decoder.



KL-Divergence

How to calculate difference between two distributions?





KL Divergence

$P(x)$

D_{KL}

$Q(x)$

WHAT

HOW

$$D_{KL}(P||Q) = \sum_{i=1}^N P(x_i) \log \frac{P(x_i)}{Q(x_i)}$$

$$D_{KL}(P||Q) \approx \frac{Q(x)}{P(x)} - 1 - \log \frac{Q(x)}{P(x)} \text{ for } x \sim P$$

Forward KL

$$D_{\text{KL}}(P\|Q) = \sum_i P(x_i) \log \frac{P(x_i)}{Q(x_i)}$$

Reverse KL

$$D_{\text{KL}}(Q\|P) = \sum_i Q(x_i) \log \frac{Q(x_i)}{P(x_i)}$$